Workshop: Categorical Analysis

your name here

This week, we are working with the Totally Serious Pop Culture Research Institute™ to answer six questions about pop culture. For every study, use $\alpha = 0.05$. Choose the appropriate categorical test, check its conditions, and interpret the result in context.

Let's call in the packages and data we need,

```
library(tidyverse)

popcorn_flavor <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M3-popcorn_flavor.csv")
streaming_genre  <-  read_csv("https://raw.githubusercontent.com/samanthaseals/
STA4173Fa26/refs/heads/main/workshops/data/M3-streaming_genre.csv")
spoiler_platform <- read_csv("https://raw.githubusercontent.com/samanthaseals/
STA4173Fa26/refs/heads/main/workshops/data/M3-spoiler_platform.csv")
costume_fandom <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M3-costume_fandom.csv")
trailer_rewatch  <-  read_csv("https://raw.githubusercontent.com/samanthaseals/
STA4173Fa26/refs/heads/main/workshops/data/M3-trailer_rewatch.csv")
meme_caption  <-  read_csv("https://raw.githubusercontent.com/samanthaseals/STA
4173Fa26/refs/heads/main/workshops/data/M3-meme_caption.csv")
```

Study 1: The Movie-Night Popcorn Debate

The institute surveyed 120 moviegoers, asking each person to choose one favorite popcorn flavor: Butter, Caramel, or Cheddar. A cinema chain claims these three flavors are equally popular. Do the survey responses provide evidence against that claim?

Dataset: `popcorn_flavor`; variable: `flavor`.

1a. Make a frequency table and report the sample percentage for each flavor.

1b. Use the appropriate hypothesis test to determine if the three flavors are equally popular.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 2: Which Meme Caption Gets Shared?

The institute randomly assigned 200 independent volunteers to see one of two captions on the same meme image: Classic or Absurdist. Each volunteer then chose whether to share the meme. Is the share proportion different between the two caption styles?

Dataset: `meme_caption`; variables: `caption_style` and `shared` (`Yes` or `No`).

2a. Construct the appropriate contingency table to display this data.

2b. Use the appropriate hypothesis test to determine if the share proportion is different between the two caption styles.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 3: Where Did You See the Spoiler?

The institute surveyed 180 fans after a television finale. Each fan named the one social platform they used most during premiere week: TikTok, Instagram, or Reddit. They also reported whether they saw a finale spoiler before watching the episode. Is spoiler exposure associated with a fan's primary platform?

Dataset: `spoiler_platform`; variables: `platform` and `saw_spoiler`.

3a. Construct the appropriate contingency table to describe this data.

3b. Use the appropriate hypothesis test to determine if spoiler exposure depends on the fan's primary platform.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 4: A Streaming Genre Prediction

A streaming service predicts that 20% of viewers will choose Comedy, 30% Drama, 20% Reality, and 30% Sci-Fi as their favorite genre. The institute independently surveyed 140 viewers, each of whom chose one genre. Do their choices follow the service's predicted distribution?

Dataset: `streaming_genre`; variable: `genre`.

4a. Make a frequency table of the observed data.

4b. Use the appropriate hypothesis test to determine if the choices follow the service's predicted distribution.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 5: The Trailer Rewatch Claim

A movie studio claims that more than 60% of viewers would rewatch its new trailer. The institute asked 120 independently selected viewers whether they would watch it again. Does the survey support the studio's claim?

Dataset: `trailer_rewatch`; variable: `rewatch` (`Yes` or `No`).

5a. Construct a frequency table to describe this data.

5b. Use the appropriate hypothesis test to determine if more than 60% of viewers would rewatch the new trailer.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...

Study 6: Convention Costume Choices

At a fan convention, the institute surveyed 180 attendees. Each person chose one primary fandom —Superhero, Fantasy, or Sci-Fi—and described their costume as Hero, Villain, or Sidekick. Is costume choice associated with primary fandom among these attendees?

Dataset: `costume_fandom`; variables: `fandom` and `costume_choice`.

6a. Construct the appropriate contingency table to describe this data.

6b. Use the appropriate hypothesis test to determine if costume choice and primary fandom are related.

Hypotheses:

- H_0 :
- H_1 :

Test Statistic and p-value

- $statistic =$
- $p =$

Rejection Region

- Reject H_0 if ...

Conclusion and Interpretation

- Reject or fail to reject H_0.
- There is or is not sufficient evidence to conclude...